JSC “Kazakh British Technical University”

School of Information technology and Engineering

SIMATIC S7-1500

SIS 2

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AUTOMATION OF TANK LEVEL CONTROL USING SIEMENS S7-1500 PLC

Introduction

Automation of technological processes increases safety, product quality, and efficiency by reducing human error and providing stable control. In this SIS project, our group designs an automation solution for a tank level control using a Siemens S7-1500 PLC. The PLC collects signals from sensors, executes control logic in real time, and sends commands to actuators. The goal is to implement reliable operation, monitoring, alarms, and safe shutdown conditions.

The Siemens SIMATIC S7-1500 PLC is well suited for tank control automation thanks to its high processing performance, advanced diagnostic capabilities, and accurate handling of analog signals. The purpose of this report is to analyze the technological process, identify the control object, define input and output signals, and justify the control structure for further implementation using a Siemens S7-1500 PLC.

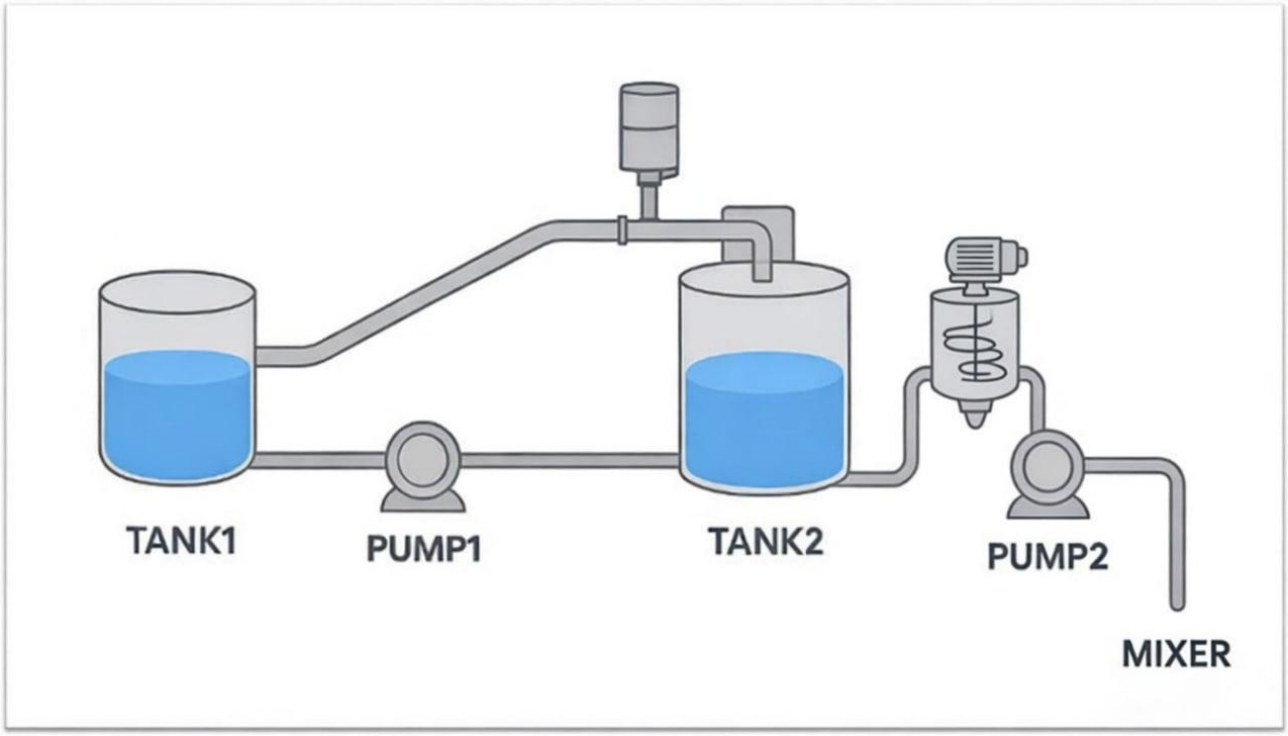
Controller: Siemens S7-1500

Programming environment: TIA Portal

Simulation: PLCSIM

1.1 Control Objective

The objective of the program is to ensure stable and safe control of a two-tank level control system with temperature protection. The PLC performs level control of Tank1 and Tank2, Pump control, Mixer control, Emergency shutdown handling, Alarm signaling.



2. Signal Table

SIS2 ▸ PLC_1 [CPU 1513F-1 PN] ▸ PLC tags

Tags | User constants | System constants

PLC tags

	Name	Tag table	Data type	Address	Retain	Acces...	Writa...	Visibl...	Supervision	Comment
1	Start	Default tag table	Bool	%I0.0		✓	✓	✓		
2	Stop	Default tag table	Bool	%I0.1		✓	✓	✓		
3	E_Stop	Default tag table	Bool	%I0.2		✓	✓	✓		
4	Lvl_Tank1	Default tag table	Bool	%I0.3		✓	✓	✓		
5	Lvl_Tank2	Default tag table	Bool	%I0.4		✓	✓	✓		
6	Temp	Default tag table	Bool	%I0.5		✓	✓	✓		
7	Pump1	Default tag table	Bool	%Q0.6		✓	✓	✓		
8	Pump2	Default tag table	Bool	%Q0.7		✓	✓	✓		
9	Mixer	Default tag table	Bool	%Q1.0		✓	✓	✓		
10	Alarm	Default tag table	Bool	%Q1.1		✓	✓	✓		
11	Run	Default tag table	Bool	%M1.2		✓	✓	✓		
12	Fault	Default tag table	Bool	%M1.3		✓	✓	✓		
13	<Add new>					✓	✓	✓		

2.1 Digital Inputs

Tag Name	Description
Start	Start pushbutton
Stop	Stop pushbutton
E_Stop	Emergency stop button
Lvl_Tank1	Level sensor Tank1
Lvl_Tank2	Level sensor Tank2
Temp	Temperature alarm

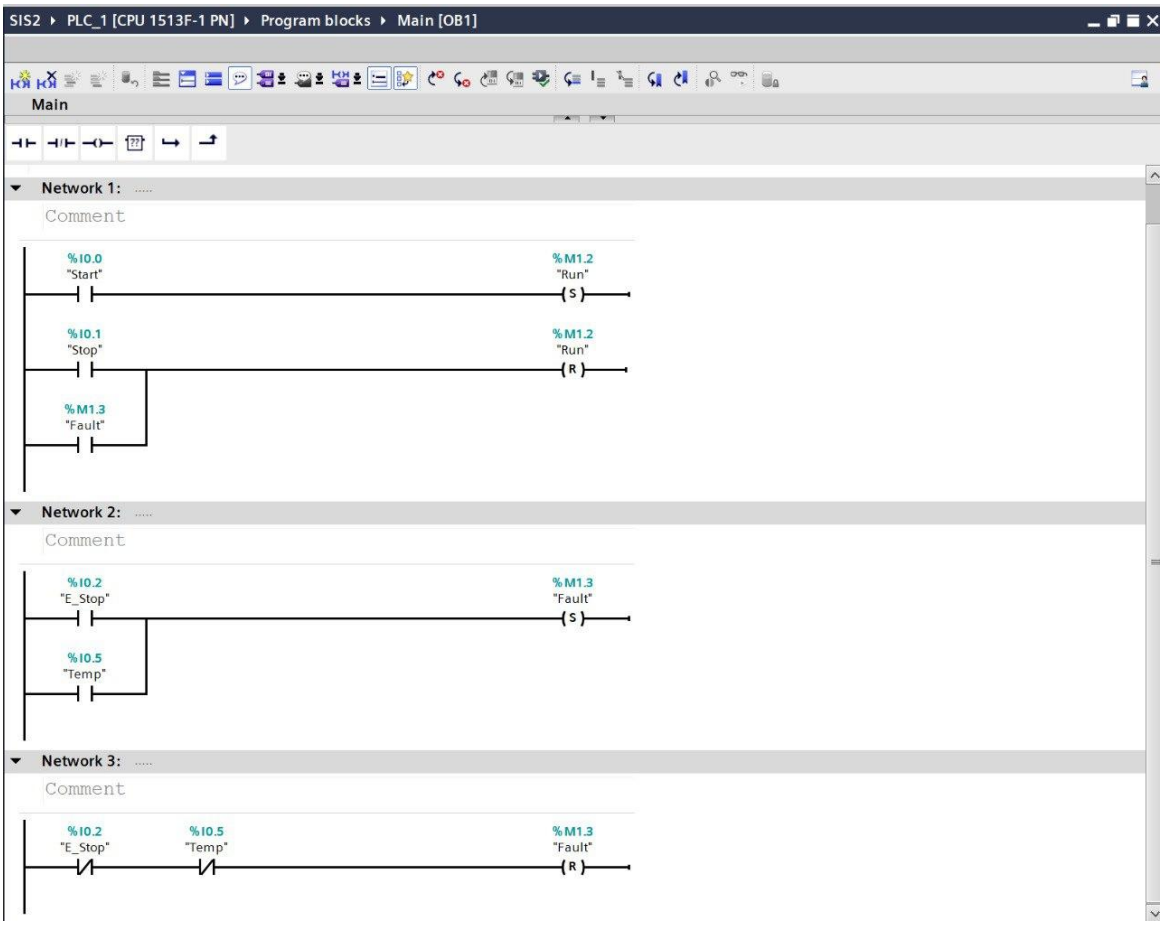
2.2 Digital Outputs

Tag Name	Description
Pump1	Transfer pump
Pump2	Cooling pump
Mixer	Mixer motor in Tank2
Alarm	Alarm signal

2.3 Internal Variables

Tag Name	Description
Run	System running
Fault	Fault condition

3. Program Logic Description



Network 1 - Run Logic

Pressing **Start** sets Run, **Stop** resets Run

If Fault, then Run = 0

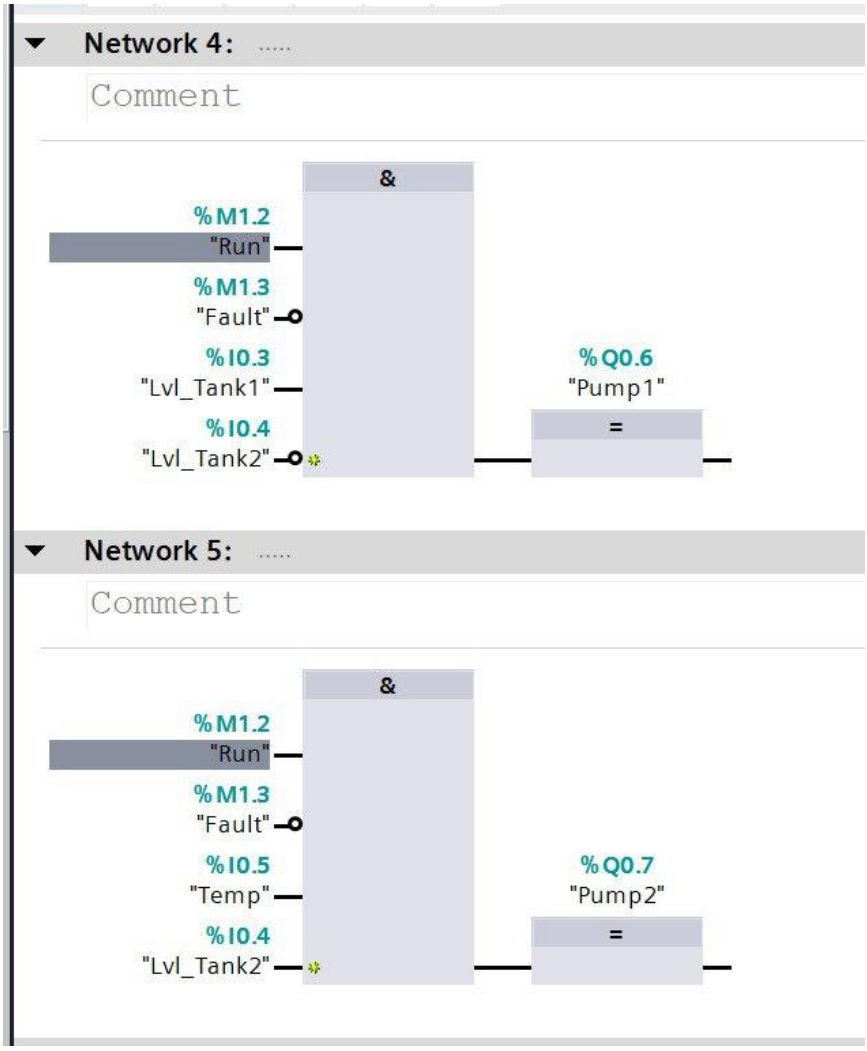
Network 2 and 3 - Fault Detection

Fault is set when:

E_Stop = 1 OR Temp = 1

Fault is reset only if:

E_Stop = 0 AND Temp = 0



Network 4 — Pump1 Control

Pump1 transfers liquid from Tank1 to Tank2.

Protection:

No dry run (Tank1 empty → Pump1 OFF)

No overflow (Tank2 full → Pump1 OFF)

No operation during fault

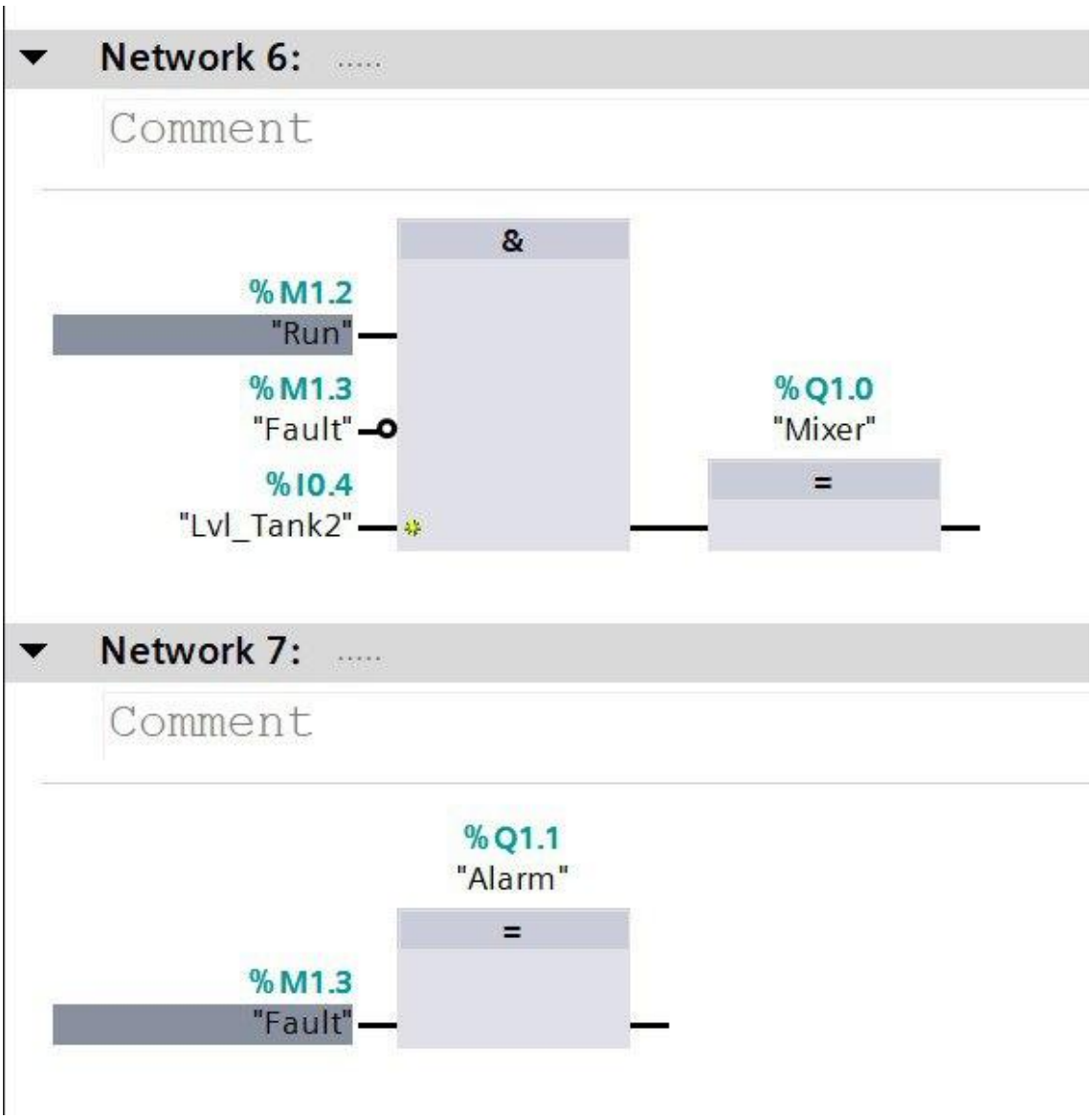
Network 5 — Pump2 Control

Pump2 operates only during overheating.

Protection:

No cooling without liquid

No operation during fault state



Network 6 — Mixer Control

Mixer works when:

Lvl_Tank2 = 1

Protection:

No dry mixing

No operation during emergency

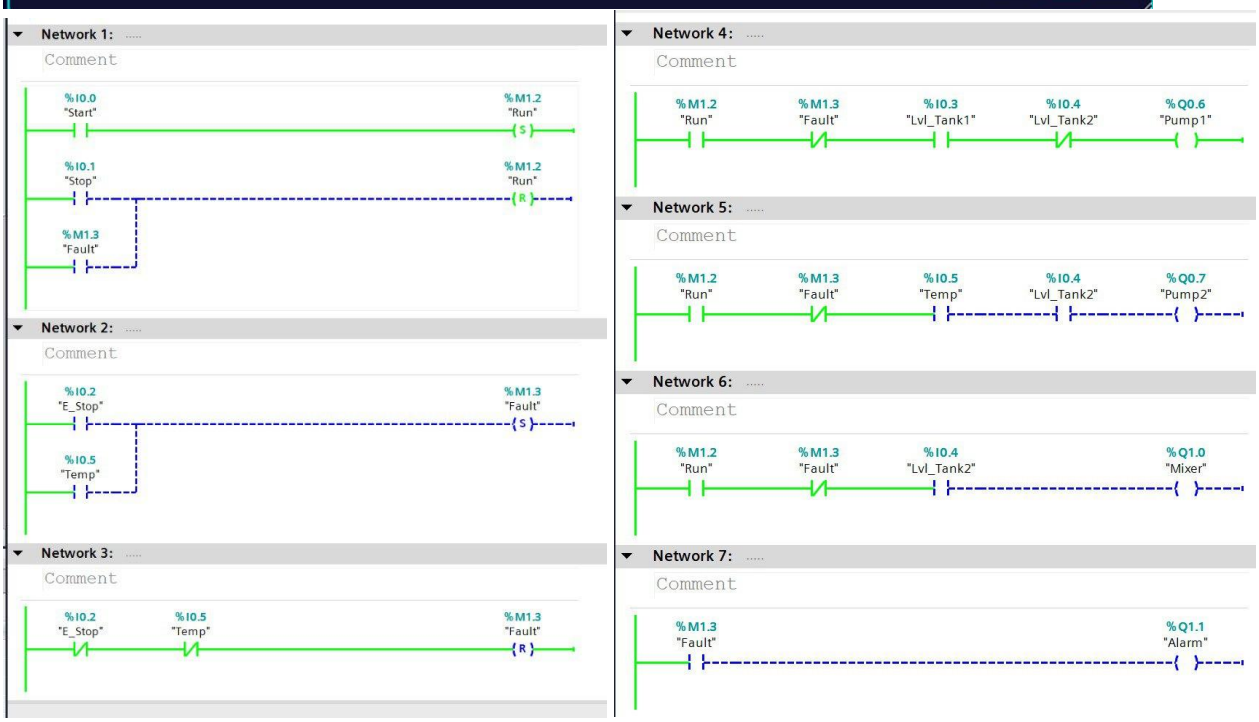
Network 7 — Alarm

Provides operator notification of abnormal state.

4. Operational Scenarios Tested in PLCSIM

Scenario 1 — Normal Start-Up

Name	Address	Display Format	Monitor/Modify State	Comment
Pump1	Q0.6	Bool	TRUE	
Pump2	Q0.7	Bool	FALSE	
Lvl_Tank1	I0.3:P	Bool	TRUE	
Lvl_Tank2	I0.4:P	Bool	FALSE	
Start	I0.0:P	Bool	TRUE	
Stop	I0.1:P	Bool	FALSE	
E_Stop	I0.2:P	Bool	FALSE	
Run	M1.2	Bool	TRUE	
Mixer	Q1.0	Bool	FALSE	
Temp	I0.5:P	Bool	FALSE	
Fault	M1.3	Bool	FALSE	
		DEC		



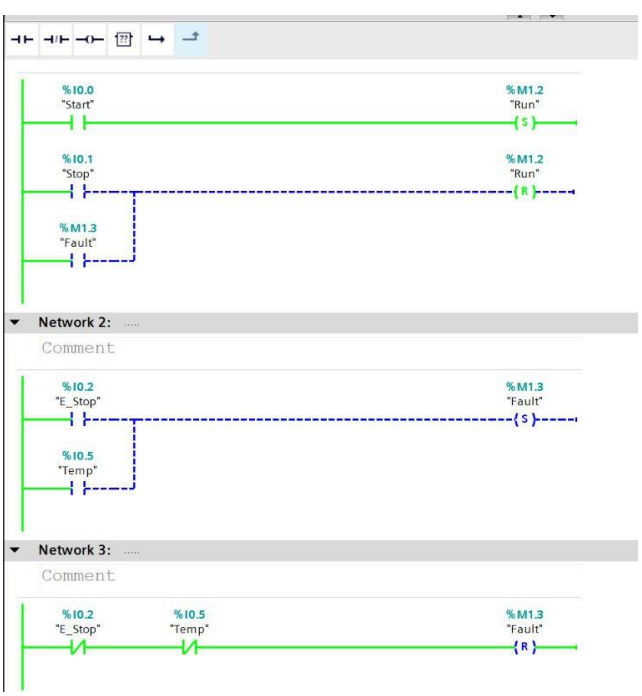
In the first scenario, at system start-up, Tank2 is empty, there are no faults, and the temperature is normal. After pressing Start, Run is set to 1, Pump1 starts, and Tank2 fills. When the tank becomes full, Pump1 stops automatically. The filling process is stable, there is no overflow, and the control logic operates correctly.

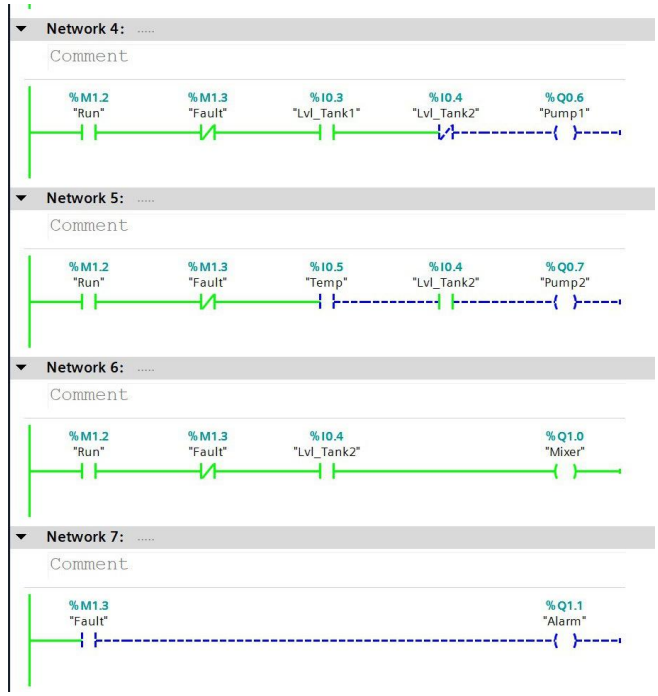
Scenario 2 — Overflow Protection

SimTable_1

PLC_1 [S7_1500]

Name	Address	Display Format	Monitor/Modify State	Comment
Pump1	Q0.6	Bool	FALSE	
Pump2	Q0.7	Bool	FALSE	
Lvl_Tank1	I0.3:P	Bool	TRUE	
Lvl_Tank2	I0.4:P	Bool	TRUE	
Start	I0.0:P	Bool	TRUE	
Stop	I0.1:P	Bool	FALSE	
E_Stop	I0.2:P	Bool	FALSE	
Run	M1.2	Bool	TRUE	
Mixer	Q1.0	Bool	TRUE	
Temp	I0.5:P	Bool	FALSE	
Fault	M1.3	Bool	FALSE	
		DEC		





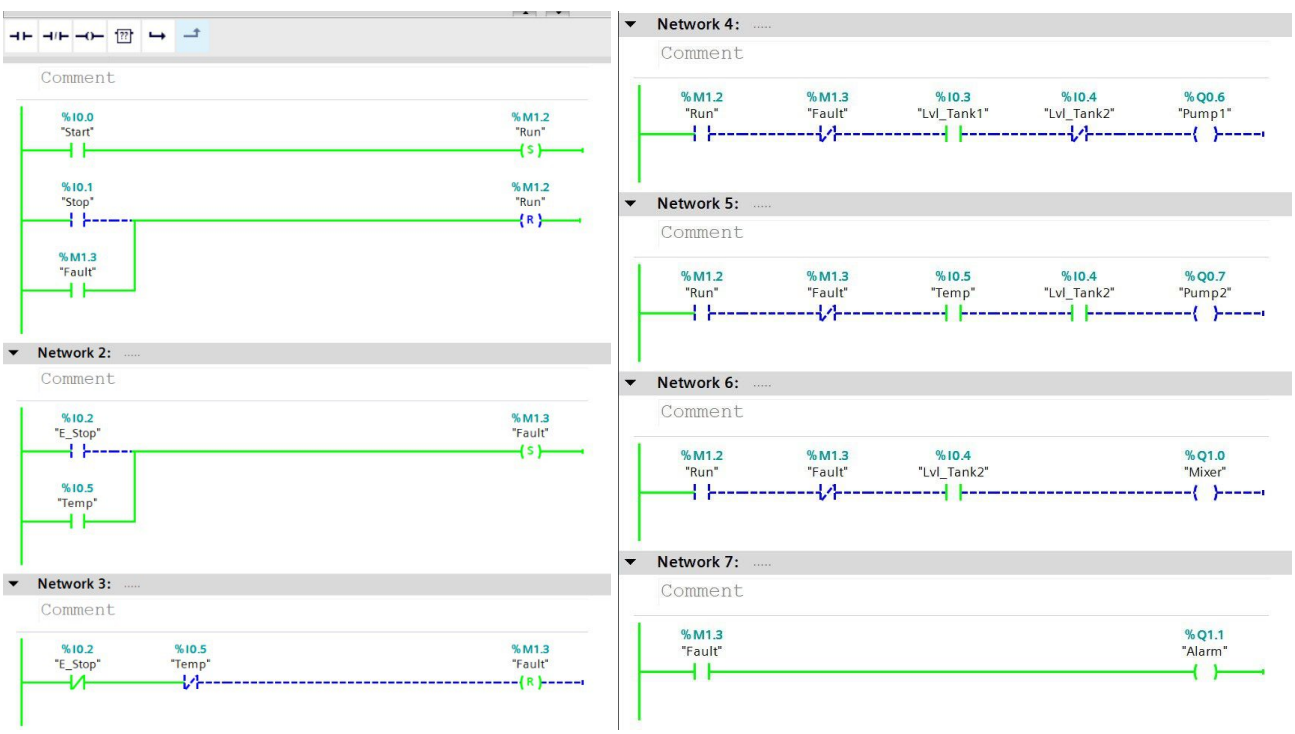
In the second scenario, Tank2 is already full. When the system is started with Run = 1, Pump1 does not start because Lvl_Tank2 = 1, preventing overflow and correctly blocking the pump.

Scenario 3 — Overheating Safe State

SimTable_1

PLC_1 [S7_1500]

Name	Address	Display Format	Monitor/Modify State	Comment
Pump1	Q0.6	Bool	FALSE	
Pump2	Q0.7	Bool	FALSE	
Lvl_Tank1	I0.3:P	Bool	TRUE	
Lvl_Tank2	I0.4:P	Bool	TRUE	
Start	I0.0:P	Bool	TRUE	
Stop	I0.1:P	Bool	FALSE	
E_Stop	I0.2:P	Bool	FALSE	
Run	M1.2	Bool	FALSE	
Mixer	Q1.0	Bool	FALSE	
Temp	I0.5:P	Bool	TRUE	
Fault	M1.3	Bool	TRUE	
		DEC		



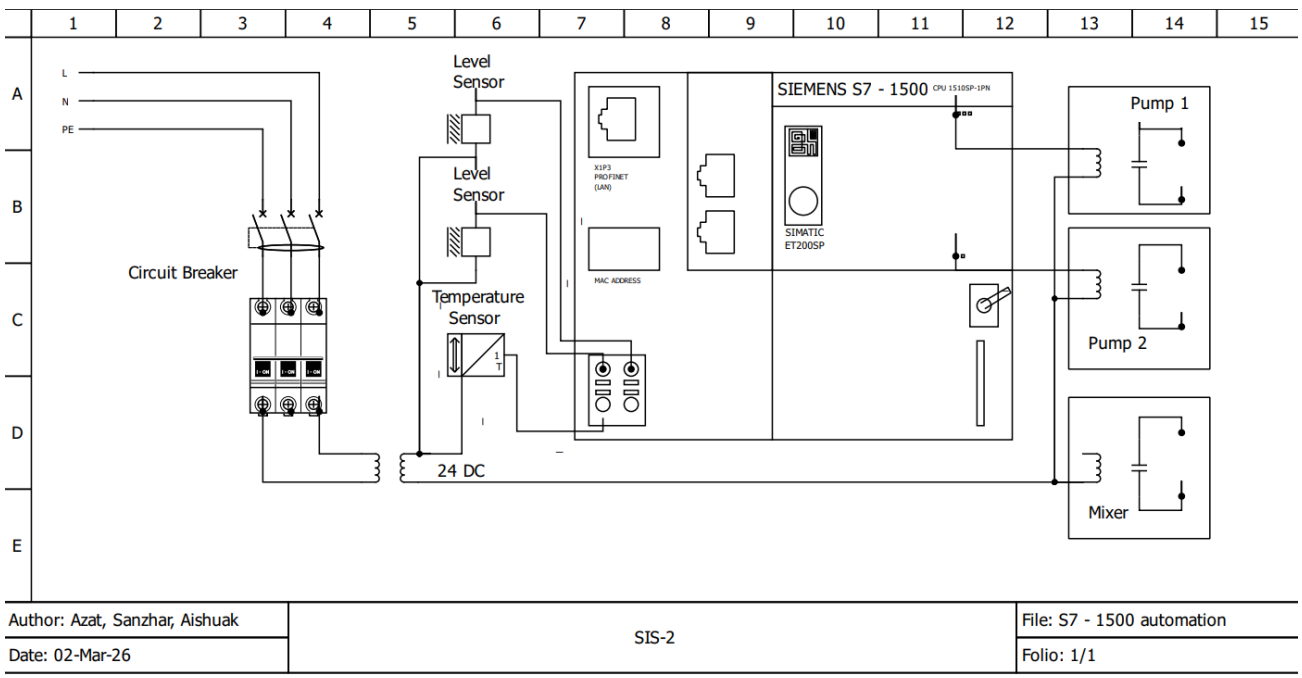
In the third scenario, while the system is running, an overheating condition occurs (Temp = 1). A fault is set, Run is automatically reset to 0, all pumps and the mixer stop, and the alarm activates. The system transitions to a safe state and remains stopped until the temperature returns to normal and the fault reset conditions are met

5. Interlocks and Safety Conditions Implemented

Implemented safety mechanisms:

- Emergency stop latch
- Overtemperature protection
- Pump dry-run prevention
- Overflow protection
- Fault state

6. Structural Diagram



The structural scheme represents the physical structure of the automation system and the interconnection of its components.